

Margaret E. B. Webb

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Education

Ph.D. Engineering Education May 2025 **Virginia Tech**, Blacksburg, VA

- *Advisor*: Dr. Marie C. Paretti
- *Dissertation Title*: Exploring Systems Influences on Engineering Graduate Student Interdisciplinary Identity-Based Motivation towards Research Careers

M.S. Civil Engineering May 2024 **Virginia Tech**, Blacksburg, VA

- *Focus Area*: Sustainable Land Development

Graduate Certificate: Disaster Resilience and Risk Management **Virginia Tech**

B.S. Mechanical Engineering: Magna Cum Laude **Rice University**, Houston, TX

Research Work Experience

Cornell University, Department of Biomedical Engineering (BME) Ithaca, NY

Postdoctoral Fellow, Cornell Disciplinary-Based Education Research (CDER) Present

- *Supervisors*: Dr. Alex Werth and Dr. Alex Coso Strong
- Conducting research on:
 - Postdoctoral professional development (PD): exploring how structures, systems, cultures, impact postdoctoral associates as scholars and change agents.
 - Generative Artificial Intelligence's role in automating qualitative assessment of student reflection quality contextualized to specific STEM Education courses.
 - Active Learning Initiatives, programs seeking to implement evidence-based teaching strategies to innovate in education based on research, as instructional change models
- Supporting research grant writing.
- Planning and coordinating an international Disciplinary-Based Education Research (DBER) conference.
- Mentoring graduate students and undergraduate students in conducting STEM Education Research.

Virginia Tech, Office of the Dean of the College of Engineering Blacksburg, VA

Graduate Research Assistant for Graduate & Professional Studies June 2024 – May 2025

Supervisor: Dr. Holly Matusovich

- Coordinated and hosted Virginia Tech Dissertation Institute (VTDI) for graduate student writing and research skills support.
- Organized College of Engineering Graduate Student Fall Welcome Fair for student orientation and networking.

- Developed and managed 2024-2025 workshop series for professional development, targeting graduate students and external fellowship holders.
- Evaluated programming effectiveness and prepared formative and summative assessment reports for the Dean.
- Initiated and developed a college-level Canvas Site as a repository for engineering graduate student professional development materials and programming.
- Developed a peer-mentoring program for external fellowship-holding students, piloting a GPS Student Ambassador Program.
- Served on the 2024 search committee for a new Assistant Director of Graduate and Professional Studies, contributing to position description writing, application screening, interviewing, and evaluation of candidates and the COE search process.
- Organized hire for a computer scientist graduate assistant.
- Collaborated in developing a scalable and efficient computerized process for addressing COE under-enrolled classes for semesterly audits and reporting.

Virginia Tech, *Institute for Creativity, Arts, and Design*

Blacksburg, VA

Designing Technology for Hospital Education, Graduate Research Assistant 2024- 2025

- Supported VT Institute for Creativity, Arts, and Technology (ICAT), Virginia Tech Major SEAD Research Grant (\$40,000) *PIs: Dr. Finda Williams and Dr. Sweta Baniya*
- *Research Data Collection and Analysis:* Conducted semi-structured interviews with hospital educators, developed a qualitative codebook, and analyzed qualitative journal and interview data using qualitative research methods.
- *Research Writing:* Published and writing conference papers and journal articles related to this research.

Virginia Tech, *Engineering Education*

Blacksburg, VA

Disaster Resilience and Risk Management, Graduate Research Assistant 2021 –2024

- Supported NSF Award # 1735139: Research Traineeship Program: Creating quantitative decision-making frameworks for multi-dimensional and multi-scale analysis of hazard impact. *Supervisor: Dr. Marie Paretti; PI: Dr. Robert Weiss*
- *Program Development and Assessment:* Supported the design of and conducted program-level and course-level assessments (including pre/post concept maps and annual interviews with student participants); summarized assessment findings for the annual NSF NRT report and Virginia Tech, presented findings to the DRRM advisory board and the graduate school.
- *Research Data Collection and Analysis:* Conducted semi-structured interviews and focus groups with participants, developed a qualitative codebook, and analyzed qualitative journal and interview data using qualitative research methods.
- *Research Writing:* Published conference papers and journal articles related to this research.
- *Conference Planning and Coordination:* Planned and coordinated the graduate student programming for the annual NRT (NSF Research Traineeship) conference for 2022-2023.
- Mentored junior Engineering Education graduate students in research methods, paper, and dissertation ideation, completion of PhD milestones, and career development, including navigating the job market.

Virginia Tech (VT), Center for Coastal Studies

Blacksburg, VA

***COVID-19 & Climate Change: Mapping the effects of compounding Crisis in the Coastal Zone using Natural Language Processing, Graduate Research Assistant* 2021 – 2024**

PIs: Dr. Andrew Katz and Dr. Sweta Baniya

Center for Coastal Studies, Virginia Tech Major SEAD Research Grant

- Supported development of an end-to-end method leveraging GPT-3.5 for generating qualitative codebooks from disaster-related social media data with minimal human input.
- Contributed to interdisciplinary research using NLP for qualitative analysis, demonstrating the potential of transformer-based large language models and generative AI for scaling up and expediting qualitative analysis of social media conversations around compounding crises.
- Mentored undergraduate students from Urban Planning, Geography, Communications, and Computer Science in applying natural language processing to interdisciplinary research on disaster resilience, guiding them in research writing and exploring technological solutions for compounding crises like COVID-19 and climate change.

VT, Center for Refugee, Migrant, and Displacement Studies

Blacksburg, VA

Towards Digital Justice: Developing a US Citizenship App & Website with Refugees & Immigrants, GRA

2021 – 2024

PIs: Dr. Sweta Baniya and Dr. Katrina Powell

American Council of Learned Societies (ACLS) Digital Justice Seed Grant

- Supported a community-engaged, participatory action research project aimed at reimagining the U.S. naturalization process for people seeking refuge through a lens of digital justice, meaningful integration, and centering community expertise.
- Employed theoretical frameworks of knowledge justice, digital justice, and epistemic integration to critically examine the unintended consequences and structural barriers created by the integration of information and communication technologies (ICTs) into the naturalization process.
- Investigated the normative assumptions about migrants and integration embedded in digitized naturalization processes and resources, and explored the knowledge, experiences, and counter-expertise of people seeking refuge that are not represented in current system designs.
- Adopted a qualitative, community-based approach to mitigate the systemic negative effects of ICT implementation in the naturalization process and co-designed ICTs with newcomers (informed by the lived experiences and values of refugee and immigrant communities).
- Mentored undergraduate students across Computer Science, Psychology, Technical Communications, and Engineering in technology development, participatory action research, and publication strategies, fostering interdisciplinary collaboration.

VT, Center for Refugee, Migrant, and Displacement Studies

Blacksburg, VA

Strengthening Virginia's Ability to Provide Migration Support and Improve Refugee Population Outcomes: Collating Refugee Resources and Collaborative Initiatives at Virginia Higher Education Institutions, GRA

PIs: Dr. Sweta Baniya and Dr. Katrina Powell

4-VA Collaborative Research Grant in collaboration with JMU, GMU, and VCU

- Supported the development of a centralized, sustainable, and public database and website that builds upon JMU's CISR's Global CWD Repository, a free resource to consolidate information on Resettlement Services for New Americans in Virginia

VT, Center for Refugee, Migrant, and Displacement Studies

Blacksburg, VA

Critical Displacement Studies Fellow

Aug 2021 – May 2024

PIs: Dr. Katrina Powell and Dr. Rebecca Hester

National Endowment for the Humanities in partnership with the Social Science Research Council

- Collaboratively wrote an open-access reader, the Interdisciplinary and Critical Displacement Studies Reader, that brings together a breadth of perspectives across disciplines and a variety of forms, building a Critical Displacement Studies methodological research approach.

Rice University, Mechanical Engineering

Houston, TX

Rice Tribomechadynamics Lab, Undergraduate Research Assistant

2016 - 2019

- Designed and fabricated fixtures for Dr. Brakes Tribomechadynamics lab, including tensile testing tools, trunnions, and drop testing equipment
- Conducted surface dimpling tests and data analysis in support of a PhD students' dissertation
- Coordinated the onboarding of new undergraduate research assistants into a research environment

Teaching Experience

Graduate Teaching Assistant Intercultural Issues in Professional Writing ENGL 3834

Virginia Tech, Blacksburg, VA

Spring 2024, Fall 2025

- Taught modules on participatory action and community-engaged technical communication, effective presentations for conferences
- Graded weekly discussion posts and course mid-semester projects

Graduate Teaching Assistant Interdisciplinary Research Methods in Disaster Resilience GRAD 5134

2023-2024

- Contributed to course design, syllabus development, and assessment development for interdisciplinary undergraduate and graduate courses.
- Responsible for grading weekly assignments and project assignments, providing detailed feedback to 25 students.
- Collaborated with faculty instructors on iterative improvements to course materials and teaching methods based on assessment data.

Middle and High School Math Teacher

Harmony Public Schools

12/2020 – 6/2021

- Designed lesson plans with active learning pedagogies for in person, online, and
- Advanced student learning by contextualizing subject matter to student interests
- Cultivated relationships with 120 students, tens of faculty, and parents
- Collaborated with teaching teams to increase the use of active learning pedagogies in the math department

Texas Education Agency, Teaching Certification Program

5/2021

- TExES 274: Grades 6-12 Mathematics, Physical Science, and Engineering Content Exam (276/300)

Solids Mechanics Undergrad Learning Assistant, Mechanical Engineering Department

Rice University, Houston, TX

Fall 2017

- Responsible for supporting recitation sessions.

- Collaborated with faculty instructors on improvements to lab materials.

Publications

Peer-Reviewed Conference Papers

1. Deters, J. R., Menon, M., Paretti, M. C., & **Webb, M.E.B.** (2021). Imagining the future interdisciplinary scholar: Exploring interdisciplinary identity development using possible selves. *Proceedings of the joint Research in Engineering Education Symposium (REES) & Australasian Association for Engineering Education Conference, Perth, Australia* [virtual].
2. **Webb, M.E.B.**, Deters, J.R., Menon, M., & Paretti, M.C. (2022). Building a Sustainable University-Wide Interdisciplinary Graduate Program to Address Disasters. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Milwaukee, Wisconsin*.
3. **Webb, M.E.B.** & Paretti, M.C. (2022). Exploring Professional Identity Development Research on Displaced Higher Education Students. *Proceedings of the Frontiers in Education (FIE) Conference, Uppsala, Sweden*.
4. Deters, J.R., **Webb, M.E.B.**, & Menon, M. (2022). Exploring Mechanical Engineering Student Perspectives of Academic Integrity During COVID-19. *Proceedings of the Australasian Association for Engineering Education Conference, Western Sydney University*.
5. Menon, M., **Webb, M.E.B.**, Josiam, M., & Deters, J.R. (2023) Sustainable Development Goals in Engineering Curriculum: A Document Analysis on Discourse and Suggestions for Improvement. *Proceedings of the Engineering Education for Sustainable Development (EESD) Conference, Colorado*.
6. **Webb, M.E.B.** & Paretti, M.C. (2023). Influences on Displaced Engineering Student Professional Identity Development: A Scoping Literature Review Across Forced Migration Contexts. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland*.
7. **Webb, M.E.B.** & Paretti, M.C. (2023). Investigating Graduate Students' Perspectives of Influences on Interdisciplinary Scholar Identity Development: An Ecological Systems Theory Approach. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland*.
8. Menon, M., **Webb, M.E.B.**, & Paretti, M.C. (2023). Understanding Faculty Perspectives of Interdisciplinary Graduate Programs. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland*.
9. Bose, B., Brotherton, H.B., Hopper, T., Oquendo, P.M., Wang, L.M., **M.E.B. Webb**, & Wilkinson, H. (2023). Graduate Student and Postdoctoral Fellow Perspectives on Advancing Women and Gender Equity in Engineering - for the Next 130 Years. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland*.
10. Scott, L., **Webb, M.E.B.**, Baniya, S., Powell, K., Salem, A. (2023). Towards Digital Justice: Developing US Citizenship Application & Website with Refugees & Immigrants. *Proceedings of the 2023 Conference on Community Writing in Denver, Colorado*.
11. Scott, L., **Webb, M.E.B.**, Baniya, S., Schell, E., Powell, K., Salem, A. (2024). Expanding Writing Abundance in Partnership with the Community. *Proceedings of the 2024 Conference on College Composition and Communication in Spokane, Washington: Writing Abundance: Celebrating 75 Years of Conversations about Rhetoric, Composition, Technical Communication, and Literacy*.
12. **Webb, M.E.B.**, Feng, X., Aarnio, H., Sundman, J., Bilow, F., Taka, M., Paretti, M., Keskinen, M. (2024). Finding Common Ground: Comparing Engineering and Design Graduate Students' Conceptualizations of Interdisciplinary Education across Two Institutions.

Proceedings of the 2024 American Society of Engineering Education (ASEE) Conference, Portland, Oregon.

13. **Webb, M.E.B.** & Paretti, M.C. (2024). Understanding Ecosystems of Interdisciplinary Graduate Education through an Ecological Systems Approach. *Proceedings of the 2024 American Society of Engineering Education (ASEE) Conference, Portland, Oregon.* ****Won ASEE 2024 Graduate Studies Division Best Student Paper Award****
14. **Webb, M.E.B.**, Baniya, S., Powell, K., El Kadima, M., Drumgold-Smith, M., & Rissler, G. (2024). Towards Digital Justice and Consolidated Resettlement Resources for New Americans: A Virginia Database Initiative. *Proceedings of the 2024 ARNOVA Conference, Washington, DC.*
15. **Webb, M.E.B.** & Paretti, M.C. (2025). Tracking the Evolution of Interdisciplinary Identity-Based Motivation in Engineering Graduate Students: A Longitudinal Case Study. *Proceedings of the 2025 American Society of Engineering Education (ASEE) Conference, Portland, Oregon.* ****Won ASEE 2025 Graduate Studies Division Best Student Paper Award****
16. Forthcoming: Smith, A., **Webb, M.E.B.**, Rasberry, N., Baniya, S., & Ogbonnaya-Ogburu, I. F. (2025). Designing Technology to Support the Hospital Classroom. In Proceedings of the 2025 CHI conference on Human factors in computing systems.

Peer-Reviewed Journal Articles

1. **Webb, M.E.B.**, Singh, H., Inman, R., Baniya, S. & Katz, A. (2025). An End-to-End Method for Inductively Generating a Qualitative Codebook with Minimal Human Input: A Comparative Study of GPT-3.5 and Traditional Codebook Generation for Scaling up Qualitative Research on Interconnected Crises. *IEEE Transactions on Technology and Society.*
2. Journal Article Accepted: **Webb, M.E.B.**; Baniya, S.; Smith, A.; Williams, I.F. (March 2026). Understanding Virginia, U.S. Hospital Educators' Work Practices and Technology Needs: A Participatory Qualitative Case Study Exploring AI Integration Opportunities. *IEEE Transactions on Professional Communication.*
3. Journal Article in Revision: **Webb, M.E.B.** & Paretti, M. (estimated 2025). Exploring Ecological Systems Influences on Engineering Graduate Student Interdisciplinary Scholar Development. *European Journal of Engineering Education Special Issue: Interdisciplinary Learning and Transformation of Engineering Education.*
4. Journal Article In Revision: **Webb, M.E.B.** & Paretti, M. (estimated 2026). Beyond Barriers and Supports: A Systems Interaction Analysis of How Microsystems Work Together to Shape Interdisciplinary Development in Engineering Graduate Education. *Journal of Engineering Education Special Issue on Surrounding Systems' Influences on Engineering Education: Focus on Organizations, Policies, and Contexts.*
5. Journal Article In Revision: **Webb M.E.B.**, Baniya S, Powell K, Rizvi M, Scott L, Drumgold-Sermen M, et al. (estimated 2026). Towards Digital Justice and Consolidated Resettlement Resources for New Americans: Reflections on A Multi-Institutional Virginia Database Initiative. *Computers and Composition.*

Book Chapters

1. Baniya, S., Powell, K., Salem, A., **Webb, M.**, Scott, L. (2025). Incorporating Community Knowledge in Design: A Reflective Account of Designing Technology with Justice. In J. Tham & J. Zhang (Eds). *Designing for Social Justice: Community-engaged Approaches to Technical and Professional Communication. Routledge ATTW Book Series.*
2. In Revisions: Baniya, S., Powell, K., Salem, A., **Webb, M.**, Scott, L., Williams, C., & East, A. (2025). Enacting Digital Justice in Partnership with Displaced Refugee and Immigrant Communities. In Powell, K, et.al., (Eds) *Critical Displacement Studies Reader. Virginia Tech Publishing.*
3. In Revisions: Feng, X., Aarnio, H., **Webb, M.E.B.**, Schiebel, J., Hingle, A. (estimated 2026). The Hidden Curriculum of a PhD: From Getting Started To Getting It Done in Engineering Education Research. *International Handbook of Engineering Education Research.*

Blog Posts

1. **Webb, M.E.B.**, Menon, M., Deters, J.R., & Paretto, M.C. (2024). Breaking Silos: Rethinking University Structures to Facilitate Interdisciplinary Engineering Graduate Education. *European Journal of Engineering Education Special Issue: Interdisciplinary Learning and Transformation of Engineering Education.*

Presentations

Invited Talks and Workshops

1. **Webb, M.E.B.** (2020, March). “Engineering Professional Development Workshop: Time Management Skills, Professional Communication, and the Job Hunt.” Invited talk by Dr. Eleazar Marquez and the Mechanical Engineering Department at Rice University in Houston, TX. *30 attendees.*
2. **Webb, M.E.B.** (2022, Spring). “Engineering Work Abroad” and “Culture in Engineering.” Invited talks by Dr. Jessica Deters and Dr. Homero Murzi at Virginia Tech Rising Sophomore Abroad (RSAP) Program. *40 attendees.*
3. **Webb, M.E.B.** & Singh, H. (2022, September). “Diversity Equity and Inclusion Research in Natural Hazards Engineering.” Invited talk with Dr. Robin Nelson and the Natural Hazards Engineering Research Infrastructure’s (NHERI) Graduate Student Council. *70 attendees.*
4. **Webb, M.E.B.** & Lightner, T. (2022, October). “Communicating Interdisciplinary Research: Taking Convergent Research Topics to Research Questions.” Invited talk with Dr. Robert Weiss and Dr. Marie Paretto Virginia Tech Disaster Resilience and Risk Management Program. *20 attendees.*
5. **Webb, M.E.B.** (2022, October). “Exploring the Professional Identity Development of Displaced Engineering Students.” Invited talk at Ohio State University for ENGE Graduate Student Seminar Exchange. <https://eed.osu.edu/events/2022/11/seminar-series-two-one-virginia-tech-eed-phd-candidates>. *30 attendees.*
6. **Webb, M.E.B.**, Singh, H, Inman, R., Katz, A., & Baniya, S. (2022, November). “COVID-19 & Climate Change: Methods for Mapping the Effects of a Compounding Crisis in the Coastal Zone” Invited talk with Dr. Robert Weiss and Virginia Tech Disaster Resilience and Risk Management Program and Center for Coastal Studies. *15 attendees.*
7. **Webb, M.E.B.** & Singh, H. (2023, February). “Cultural Competence in Hazards and Disaster Research: CONVERGE Training Module Workshop.” Invited talk with Dr. Robin Nelson and

- the Natural Hazards Engineering Research Infrastructure's (NHERI) Graduate Student Council. <https://www.youtube.com/watch?v=a4MKgKWjRjM&t=52s>. *50 attendees*.
8. **Webb, M.E.B.**, Baniya, S., East, A., Williams, C., & Powell, K. (2023, February). "Towards Digital Justice: Participatory Action Research on Structural Barriers to US Citizenship for People Seeking Refuge." Invited talk with Dr. Janine Joseph at Virginia Tech Introduction to Displacement Studies Program. *25 attendees*.
 9. **Webb, M.E.B.**, Baniya, S., & Powell, K. (2023, March). "Towards Digital Justice: Participatory Action Research on Structural Barriers to US Citizenship for People Seeking Refuge." Invited talk with VT Center for Refugee, Migrant, and Displacement Studies and VA CRMDS Consortium. *50 attendees*.
 10. **Webb, M.E.B.**, Gerhardt, M., & Schibelius, L. (2023, Oct). "Navigating Engineering Industry: The Job Search and Beyond." Invited talk with Virginia Tech Interdisciplinary Senior Capstone course about the transition to engineering work. *60 attendees*.
 11. **Webb, M.E.B.** (2024, September). "Planning to Apply for Engineering Graduate School Fellowships." Invited talk with the VT College of Engineering Graduate and Professional Studies program. *25 attendees*.
 12. **Webb, M.E.B.** (2025, December). "Conference and Poster Planning for Undergraduate Technical Communication Students." Invited talk and poster judge with Dr. Baniya's undergraduate Technical Communication and Documentation course. *30 attendees*.
 13. **Webb, M.E.B.** and **Smith, A.** (2025, February). "Understanding Virginia Hospital Educators' Work Practices and Technology Needs." Invited talk with VT Institute for Creativity, Arts, and Technology. *30 attendees*.
 14. **Webb, M.E.B.** (2025, April). "Poster Presentations for Research Conferences." Invited talk with the Virginia Tech College of Engineering Graduate and Professional Studies (COEGPS). *25 attendees*.
 15. **Webb, M.E.B.** (2025, May). "Getting the Most Out of Engineering Conferences and Conference Resources." Invited talk with the Virginia Tech College of Engineering Graduate and Professional Studies (COEGPS). *25 attendees*.

Conference Presentations and Posters

1. **Webb, M.E.B.**, Schwarz, B., Sobti, S., King, P., Montalbano, B., Broadman, C., & Ghorbel, F. (2019, May). "BuoyBots: A Soft-Robotics Approach to Buoyancy Control." Presented at the Rice University Oshman Engineering Design Conference.
2. **Webb, M.E.B.**, Fraser, R., & Davies, H.R. (2022, March). "Psychosocial Factors in Disaster Research." Hosted a Roundtable Discussion at PRIMR Conference at the University of North Texas. *10 attendees*.
3. Fiore, S., Hartley, T., Granados, D., & **Webb, M.E.B.** (2022, August). Panel Session: Understanding and Addressing Interdisciplinary Learning: Perspectives from NSF Funded Research to Improve Team Science Education. Presented at the International Network for the Science of Team Science (SciTS) Conference [virtual].
4. **Webb, M.E.B.** (2022, October). "Facilitating Interdisciplinary Graduate Education in Disaster Resilience" NSF Research Traineeship Annual Meeting. *100 attendees*.
5. **Webb, M.E.B.** (2022, October). "Turning Convergence Research Topics into Convergence Research Questions and Teams" NSF Research Traineeship Annual Meeting. *70 attendees*.
6. Davies, H.R., **Webb, M.E.B.**, & Fraser, R. (2023, January). "Psychosocial Factors and Perishable Data Collection in Disasters: Creating Resilience through Research." Hosted a Roundtable Discussion at the Society for Social Work Research (SSWR) Conference. *10*

attendees.

7. **Webb, M.E.B.,** Scott, L., Salem, A., Baniya, S., & Powell, K. (2023). Towards Digital Justice: Developing US Citizenship Application & Website with Refugees & Immigrants Using Participatory Action Methods. Proceedings of the Conference on Community Writing Workshop, Denver, Colorado.

Recognition

Honors and Awards

SEFI Early Career Researcher's Club: European Society of Engineering Education '25-'26

American Society of Engineering Education Graduate Studies Division (GSD) June 2025

Best Student Paper Award GSD, second year in a row

First author for "Tracking the Evolution of Interdisciplinary Identity-Based Motivation in Engineering Graduate Students: A Longitudinal Study" with co-author Dr. Marie C. Paretti

<https://sites.asee.org/gsd/division-awards/>

American Society of Engineering Education Graduate Studies Division (GSD) June 2024

Best Student Paper Award GSD

First author for "Understanding Ecosystems of Interdisciplinary Graduate Education through an Ecological Systems Approach" with co-author Dr. Marie C. Paretti

Virginia Tech

Nominated for VT COE Outstanding Doctoral Student 2025

1st place VT COE Torgersen Outstanding Dissertation Awards May 2025

<https://news.vt.edu/articles/2025/05/>

<https://torgersengradaward.vt.domains/2025-winners/>

President's Honor Roll 2021-Present

Pratt Fellowship for Research Excellence, \$2,000 2021-2022

Outstanding Capstone Design Project:

Bowman Sustainable Land Development CEE M.S. Program Spring 2023

<https://liddi.cee.vt.edu/content/>

Rice University 2015-2019

President's Honor Roll

Rice University Willy Revolution Award for Outstanding Innovation: Senior Capstone Award

<https://oedk.rice.edu/Sys/PublicProfile/47555555/1063096>

Oshman Engineering Design Engineering + Art Competition

<https://oedk.rice.edu/news/7213404>

Engineering Work Experience

Subsea Engineer, ExxonMobil 8/2019 – 3/2021

- Rotating Facilities Engineer to Norman Wells in NWT, Canada: Responsible for on-the-

ground engineering activity and operations optimization

- Project Engineer Liza Phase 1 Subsea Pig Launcher: Developed, manufactured, and tested a \$3M assembly that was deployed offshore Guyana for a remediation activity
- Subsea Hardware Lead Central South Dag: Responsible for the design of all subsea hardware for a 14-well tieback offshore Sakhalin, Russia
- Internship Program Coordinator for Upstream Integrated Solutions: Planned and executed the internship program for 80 employees. Developed projects, assigned mentorship, and facilitated the development of new hires.
- Subject Matter Expert in Template Manifolds and Elastomer-Sealed Subsea Gaskets: Served as the main point of contact for technical questions and training on these two SME topics for all Exxon subsea projects.

Subsea Engineering Intern, ExxonMobil

5/2018 – 8/2018

- Designed an enhanced surveillance leak detection tool for potential methanol egress in subsea fields
- Performed a calculated risk assessment on critical path mid-water arches and recommended areas of focus for repair
- Oversaw the site receipt testing of production header isolation valves prior to assembly-level fabrication
- Scoped the 2018 cathodic protection survey for Mobil Equatorial Guinea using descriptive analytics in JMP

Process Engineering Intern, Air Liquide

5/2017 – 8/2017

- Designed an atmospheric pressure vessel that meets ASME safety requirements regarding natural gas disposal
- Collaborated with plant managers and process technicians to write new standard operating procedures for an off-site air separation facility

News

Virginia Tech News: Class of 2025 Six students graduate with Ph.D. from Department of Engineering Education

June 2025

<https://news.vt.edu/articles/2025/06/Classof2025DepartmentofEngineeringEducation.html>

Virginia Tech News Webb wins 1st place VT College of Engineering Torgersen Outstanding Dissertation Awards

May 2025

<https://news.vt.edu/articles/2025/05/eng-coe-torgersen-research-winners-2025.html>

<https://torgersengradaward.vt.domains/2025-winners/>

Cornell University NSF-funded postdocs to research education across disciplines

2024

<https://as.cornell.edu/news/nsf-funded-postdocs-research-education-across-disciplines>

Virginia Tech Institute for Creativity, Arts, and Technology SEAD Grant Stories: Weaving Weeds, Exploring human-plant relationships through bio-fabrication

<https://icat.vt.edu/projects/supported-research/sead-grants/2023-2024/student/weaving-weeds.html>

Virginia Tech College of Liberal Arts and Human Sciences News Stories <u>Internships help students understand the needs of refugees and migrants</u>	2022
Virginia Tech Bowman Sustainable Land Development Civil Engineering News Outstanding Civil Engineering Design Team: <u>Bowman Sustainable Land Development Program Holds Annual Winter Meeting</u>	2023
Reuters: “Exxon exodus turns floating 'cube' into Internet meme” mention <u>https://www.reuters.com/business/exxon-exodus-turns-floating-cube-into-internet-meme-2021-10-04/</u>	2021
Rice University Oshman Engineering Design Kitchen News <u>https://oedk.rice.edu/news/8792171</u> : Engineering Design Showcase Willy Revolution Award <u>https://oedk.rice.edu/news/7213404</u> : Engineering Art Competition	Spring 2019

Fellowships

Colorado School of Mines Teaching with Heart Program \$500 fellowship to participate in experiential workshops focused on integrating empathy and care into STEM teaching	2024-2025
Institute for Creativity, Arts, and Technology Virginia Tech Roger and Debbie West Student Research Grant, \$1000 Weaving Weeds: Exploring Human-Plant Relations Through Bio-Fabrication	2023-2024
Society for Social Work Research Doctoral Student Travel Award, \$1500	2023
American Society of Engineering Education: Women in Engineering Division Graduate Student Travel Grant, \$1500	2023
National Science Foundation in Partnership with Virginia Tech NSF Research Traineeship Fellowship Disaster Resilience	2022-2023
Virginia Tech Travel Grant, Graduate and Professional Student Senate, \$1500 Travel Grant, Center for Coastal Studies: Disaster Resilience	2022-2023 2022-2023
Rice University Rice Engineering Alumni Association Buckley Sartwelle Scholarship, \$12,000	2015-2019

Grant Writing Experience

<i>NSF STEMEdIPRF Postdoctoral Fellowship Proposal</i> Transforming Graduate Education and Systems Perspectives on Interdisciplinary STEM Graduate Development and Programming PI: Margaret Webb • Led ideation and conceptual development • Collaborated with mentors across universities • Completed proposal writing and submission	December 2024
<i>NSF CAREER Award Grant Application work for Dr. Sweta Baniya, GRA</i>	Summer 2024

Theorizing Disaster Rhetorics: A Multi-Country Study for Disaster Response and Crisis Communication

PI: Dr. Sweta Baniya

- Conducted an extensive literature review to inform the theoretical framework and support the grant proposal.
- Collaborated on writing and editing key sections of the proposal, enhancing clarity, coherence, and alignment with NSF priorities.

Service

Service Roles

Rice University Engineering Alumni Association <i>Education Research Consultant</i>	05/2025 - Present
Virginia Tech Engineering Outstanding Doctoral Student Award Committee	Spring 2024
VT Engineering Dean's Search Committee for Assistant Director Grad Studies	Fall 2024
Virginia Tech Graduate and Professional Student Senate (GPSS) <i>Representative for the Department of Engineering Education</i> <i>Voting Member of the Commission on Faculty Affairs</i>	2023-2024
American Society of Engineering Education Student Division Best Paper Chair	2023-2024
Virginia Tech Chapter of American Society of Engineering Education (ASEE@VT) <i>Vice President</i>	2022-2023
Natural Hazards Engineering Research Infrastructure Graduate Student Council (GSC) <i>Chair of DEI and President of Social Science Research Group</i>	2022 - 2024
ExxonMobil, Subsea Engineering Department <i>United Way Champion</i>	Houston, TX 2019-2020
Children's Assessment Center (CAC) <i>Volunteer Program Coordinator</i>	Houston, TX 8/2018-5/2020
Rice University, Department of Mechanical Engineering <i>Co-Founder of Mechanical Engineering Undergraduate Advisory Board</i> <i>Student Representative to the Credit Cap Committee</i> <i>Student Ambassador to Rice Engineering</i> <i>Peer Career and Peer Academic Advisor</i>	Houston, TX 2016/2019 8/2015 – 5/2016 2015 – 5/2019 6/2015-5/2019

Professional Membership

VT Community of Practice for Professional Development	12/2024 – 5/2025
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AAAS/Science Program for Excellence in Science	10/2024 - Present
American Society for Engineering Education	8/2021 – Present
Graduate Studies Division	
Multidisciplinary Engineering Division	
Student Division	
Natural Hazards Research Infrastructure Graduate Student Council	8/2021 – 5/2024
DEI Committee	
Social Science Research Committee	
Rice University Engineering Alumni Association	05/2019 - Present
Society of Petroleum Engineers	8/2021 - Present
ExxonMobil ToastMasters	8/2019 – 12/2020
ExxonMobil Black Employees Success Team	8/2019 – 12/2020
ExxonMobil PRIDE	8/2019 – 12/2020
ExxonMobil Women’s Interest Network	8/2019 – 12/2020
Society of Women Engineers	8/2015 – Present
American Society of Mechanical Engineers	8/2015 – Present
Rice Creative Society	8/2015 – 5/2019

Professional Development

Conference Attendance, Paper Reviews, Moderation

Collaborative Network for Engineering and Computing Diversity Conference New Orleans, LA <i>In-Person Attendee</i>	2022
The Natural Hazards Workshop Boulder, CO at UC Boulder CONVERGE Facility <i>Virtual Attendee</i>	Spring 2022
Frontiers in Education (FIE) Conference, Uppsala, Sweden <i>In-Person Attendee</i> <i>Reviewer</i>	Fall 2022
Australasian Association for Engineering Education (AAEE) Conference <i>Virtual Attendee</i> <i>Reviewer</i> <i>PhD Research (HDR) Symposium</i>	Winter 2021 and 2022 Winter 2022 12/2021
American Society of Engineering Education (ASEE Conference) <i>In-Person Attendee</i> <i>Reviewer</i> <i>Moderator</i>	Summer 2022 - Present
Natural Hazards Research Infrastructure Graduate Student Council Conference <i>Virtual Attendee</i>	2023

Professional Development Workshops Attended

Virginia Tech Newman Library <i>Evidence Synthesis Reviews 101</i>	Virtual 9/2021
Virginia Tech Dissertation Institute Summer 2024	In-Person
Colorado School of Mines Teaching with Heart Program	Virtual 2024-2025

Relevant Graduate Coursework

Engineering Education

- ENGE 5214: Issues in Engineering Education
- ENGE 5224: Discipline-Based Literature in Engineering Education
- ENGE 5604: Engineering Education Research Methods
- ENGE 6614: Quantitative Data Analysis in Engineering Education
- ENGE 6624: Qualitative Data Analysis in Engineering Education
- ENGE 5514: Applied Design and Assessment of Engineering Education Experiences

Higher Education and Organizational Theory

- EDHE 6304: Theories of Educational Organizations
- EDHE 6054: College and University Administration
- EDHE 5314: College Student Development Theory (in progress)

Interdisciplinary Studies and Disaster Resilience

- GRAD 5034: Interdisciplinary Disaster Resilience
- GRAD 5134: Research in Disaster Risk Reduction and Management
- CEE 5264: Sustainable Land Development
- CEE 5274: Land Development Design Projects
- MINE 5114: Sustainable Mineral and Energy Resources

Social Justice and Identity in Education

- WGS 5424: Identity, Migration, and Place
- ENGE 5304: Graduate Student Success in a Multicultural Environment

Policy and Decision-Making

- SPIA 5124: Policy Decision-Making in STEM-H

Sustainability and Resilience in Engineering

- CEE 5114: Advanced Sustainable Systems
- CEE 5004: Adaptive Reuse & Redevelopment

- CEE 5204: GIS Applications in Civil Engineering

Research and Independent Study

- ENGE 7994: Research and Dissertation: Ecological Systems of Interdisciplinary Graduate Education
- LAHS 5974: Independent Study: Critical Displacement Ethics in Civil Engineering